# SolarShare — Product Requirements Document (PRD)

**Track:** Energy and Sustainability | **Stage:** Ideation → MVP | **Version:** 1.0 | **Date:** August 2026

---

## 1. Executive Summary

SolarShare is a peer-to-peer (P2P) solar energy trading platform that lets rooftop solar owners ("prosumers") sell surplus electricity directly to nearby non-solar consumers, at rates better than the DISCOM feed-in tariff but cheaper than retail. Rather than building new grid or metering infrastructure, SolarShare is designed as a consumer-facing app layer — a **Beckn Application Provider (BAP)** — that plugs into India's own **India Energy Stack (IES)**, the Ministry of Power's live, government-backed P2P energy trading framework, first demonstrated to the Prime Minister on 16 February 2026 and currently piloting in Delhi and western Uttar Pradesh. SolarShare's

differentiation is a closed-loop, housing-society-first go-to-market that existing Trade Platform Providers (TPPs) have not specifically targeted.

## 2. Problem Statement

- India's residential rooftop solar base is scaling faster than any comparable market: **PM Surya Ghar: Muft Bijli Yojana** crossed **50.06 lakh beneficiary households by July 2026**, is targeting **75 lakh by December 2026**, and **1 crore households by FY 2026–27**, with total national solar capacity crossing **150 GW**.
- Existing net metering under-monetizes this surplus. DISCOMs buy back excess power at low feed-in tariffs. Even so, **over 12 lakh households collectively earned ₹421 crore** from selling surplus power in FY 2024–25 — real, growing demand for a better mechanism, at today's low efficiency.
- Meanwhile, non-solar households in the same neighborhood keep paying full retail tariff, even when a solar-rich neighbor is exporting unused capacity at the same moment, on the same local grid line.
- The Ministry of Power has validated this exact gap at a policy level: DERC and UPERC approved a **six-month P2P trading pilot in**

**February 2026** under the IES framework. But the pilot is infrastructure, not a product — it requires registered consumer-facing apps (BAPs) to actually reach households. SolarShare is built to be one of those apps.

## 3. Goals & Objectives

| Goal | Metric |
|---|---|
| Validate P2P trading demand in a closed community | 1 housing society pilot, 50+ active prosumer/buyer pairs |
| Give prosumers a better return than net metering | Seller rate 20–30% above feed-in tariff |
| Give buyers a cheaper alternative to retail tariff | Buyer rate 10–15% below DISCOM retail rate |
| Build trust through transparent settlement | 100% of trades logged with tamper-evident transaction IDs |
| Operate as an asset-light software layer | Zero owned hardware; uses existing smart/net meters only |

## 4. Target Users / Personas

**1. The Prosumer ("Priya")** — owns rooftop solar under PM Surya Ghar, has a smart net meter, generates more than she consumes on sunny days, currently accepts whatever the DISCOM net-metering rate offers.

**2. The Buyer ("Rahul")** — lives in the same housing society or locality, has no solar, pays full retail tariff, wants cheaper and greener electricity without installing panels.

**3. The Society Admin ("RWA Coordinator")** — manages the housing society's shared infrastructure, is the trusted local authority who can drive bulk onboarding and vouch for the platform's legitimacy to residents.

## 5. Product Scope

### In scope — v1 (Hackathon / Prototype)

- Prosumer surplus-listing screen (manual or simulated smart-meter input)
- Buyer marketplace/matching screen
- Simulated transaction ledger with tamper-evident hash IDs
- Trade history and basic dashboard

### In scope — Phase 2 (Pilot, post-hackathon)

- Real integration with a DISCOM-issued smart/net meter feed (read-only, via AMISP/Head-End System data)
- Registration as a Beckn Application Provider (BAP) on the UEI network
- WhatsApp-based onboarding and trade notifications (mirroring the real IES pilot's UX pattern)
- RBI-compliant escrow-based payment settlement

### Out of scope — v1

- Owning or deploying any hardware (meters, batteries, inverters)
- Interstate trading (requires CERC-level clearance, not needed for a single-society pilot)
- Battery storage / Virtual Power Plant features (explicitly deferred — see Section 9)

## 6. Functional Requirements (MoSCoW)

| Priority | Requirement |
| --- | --- |
| Must | Prosumer can list available surplus (kWh + asking price) |
| Must | Buyer can view available listings and initiate a purchase |
| Must | Every completed trade generates a unique, timestamped, tamper-evident record |
| Must | Trade history is viewable by both parties |
| Should | Price suggestions shown relative to DISCOM retail tariff and feed-in tariff |
| Should | WhatsApp notification on trade match/completion |
| Could | AI-based demand/surplus forecasting (see Section 9) |
| Could | Gamified sustainability score / leaderboard per society |
| Won't (v1) | Battery-backed storage trading |
| Won't (v1) | Cross-DISCOM / interstate trading |

## 7. Non-Functional Requirements

- **Compliance:** Must align with CEA Smart Meter Regulations 2023 (bidirectional, AMI-enabled metering) and applicable state DISCOM net-metering guidelines.
- **Security:** All user identities verified as genuine DISCOM customers (mirrors the IES "verified credential" model); payment settlement via RBI-compliant, escrow-backed gateway.
- **Availability:** Platform must gracefully fall back to standard grid supply when no local surplus is available — never blocks a buyer's power access.
- **Data Privacy:** Consumption/generation data handled per India's Digital Personal Data Protection (DPDP) Act, 2023.

## 8. Success Metrics (Pilot Phase)

- Number of active prosumer–buyer pairs
- Total kWh traded peer-to-peer vs. total society consumption
- Average seller premium over feed-in tariff / buyer discount vs. retail tariff
- Trade completion rate (listed vs. matched vs. settled)

- $CO_2$ emissions avoided (see Impact Metrics in Implementation Plan)

## 9. Assumptions & Constraints

- Assumes the pilot society already has, or is willing to apply for, net metering and smart meters (a growing baseline given PM Surya Ghar and RDSS rollout).
- Assumes DERC/UPERC-style pilots expand to more states within 12–18 months; SolarShare's initial addressable geography is therefore constrained to pilot-approved zones until then.
- Battery storage is explicitly out of scope for v1 — the shared DISCOM grid line acts as the buffer, keeping the model asset-light (see disadvantages/mitigation in Section 10).

## 10. Key Risks & Mitigation (Summary)

| Risk | Mitigation |
| --- | --- |
| Regulatory dependency on state pilots | Start in DERC/UPERC-approved zones; track multi-state expansion via IES Alliance |

| Risk | Mitigation |
| --- | --- |
| Low per-trade margin | Diversify revenue: transaction fee + SaaS + carbon credit layer (see PRD companion: Market doc) |
| Trust/adoption barrier | Onboard via RWA/society admin, not cold individual marketing |
| Weather-dependent supply | Automatic, transparent fallback to grid supply |

## 11. Roadmap Snapshot

See **Implementation Plan** document for the full phased timeline (Hackathon MVP → Single-society pilot → Multi-society expansion → UEI/BAP registration).